

Utkarsh Sonawane

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EDUCATION

The Pennsylvania State University

B.S. Computer Science | Capital Honors Program

Harrisburg, PA

Expected Dec 2028

- **GPA:** 3.53/4.00 | Dean's List
- **Coursework:** Data Structures & Algorithms; Object-Oriented Programming and Calculus III (in progress)

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, Swift, SQL

Backend & Data: FastAPI, Node.js, Express, PostgreSQL, Supabase, REST APIs, pgvector

Tools & Clients: Git, Linux, React, Next.js, SwiftUI, HealthKit, ESP32 (Arduino/C++)

PROJECTS

Nytr — iOS Nutrition & Training System | *Python, FastAPI, PostgreSQL, SwiftUI*

2026

- Built a production-validated iOS and FastAPI/PostgreSQL system integrating Stacks Market dining data, HealthKit, Hevy, and barcode food records into daily nutrition and training guidance.
- Engineered an auditable planning backend with PostgreSQL row-level security, immutable versioned evidence, append-only corrections, and exact Decimal arithmetic to prevent incomplete data from becoming unsupported recommendations.
- Validated meal planning, goal tracking, and training-coaching workflows with 1,000+ backend tests, 250+ iOS tests, and physical-device testing.

Devvy — Local Developer IPC Service | *TypeScript, Node.js, Unix Sockets*

2026

- Built a local Discord presence daemon supporting VS Code, OpenCode, and Command Code on macOS and Windows through structured localhost events and direct Discord IPC.
- Unified 3 competing editor/agent integrations into a single privacy-conscious activity feed using priority arbitration, heartbeat expiration, binary IPC framing, and reconnect handling.

ScholarAI — Government Scheme Discovery | *Python, PostgreSQL, pgvector*

2026

- Built the data ingestion pipeline for a 5-person hackathon team, normalizing 1,000+ scraped government schemes into PostgreSQL through batched inserts.
- Generated embeddings and loaded scheme records into pgvector for semantic retrieval; the team advanced to the final round of the USAII Global AI Hackathon (6,000+ registrants).

Clage — NEAT Engine & Artificial Life | *Python, Algorithms*

2025 – 2026

- Implemented NEAT from scratch in Python, including mutation, innovation tracking, crossover, and speciation; evaluated the engine on 4 benchmark tasks (OR, AND, XOR, sine).
- Built a 2D artificial-life simulation and reproducible experiment harness with fixed-seed benchmark runs and one-factor-at-a-time environmental sweeps.

LEADERSHIP & ACTIVITIES

Technical Chair, IEEE

IEEE Student Branch, Penn State Harrisburg

Harrisburg, PA

Sep 2025 – Present

- Hosted 2 technical workshops on Markdown and building with Google Gemini, translating developer tools and applied AI workflows into hands-on student sessions.
- Elected Technical Chair in Sep 2026 to coordinate software, AI, and robotics workshops and projects for the IEEE student branch.

Founder & Editor-in-Chief

Glimpse — Student Publication

Dubai, UAE

Jun 2024 – Feb 2025

- Grew a student publication from launch to 250+ subscribers by authoring and editing 30+ articles on technology, science, and student life.

HONORS & CERTIFICATIONS

Honors: USAII Global AI Hackathon finalist | Capital Honors Program | Dean's List

Certifications: Ethics of AI (*University of Helsinki*) | AI Essentials (*Penn State*)